

Plant Emotion Detector: Do Plants Detect Human Emotions?

COIN-Seminar 2026 - Group 1

Nils Bringewatt, 7443820

Jakob Jetter, 7443798

Sarah Löhnert, 7380297

Elisabeth Stein, 7443682

Supervisor: Prof. Dr. Peter Gloor

Co-Supervisor: Simon Wolf

Department of Information Systems for Sustainable Society, Faculty of Management, Economics and Social Sciences, University of Cologne

Abstract This paper investigates whether a plant's bioelectrical voltage signal carries information about the emotional state of a nearby human. In a multimodal experiment, 15 participants each watched a fixed sequence of emotion-eliciting video clips while a *Kalanchoe* plant's voltage, the participant's heart-rate variability (HRV), facial expression, and ambient CO₂ were recorded. Emotion was operationalized along the circumplex dimensions of valence (from facial expression via HSEmotion) and arousal (from HRV), yielding proxy labels for 284 thirty-second windows. Three classifiers, Random Forest, XGBoost, and Neural Network, were trained to predict valence, arousal, and a combined emotion label from plant-derived features alone, using participant-grouped nested cross-validation and permutation testing. The plant signal predicted valence above chance (macro-F1 = 0.42, $p = 0.005$). The emotion label was statistically significant but weak in absolute terms (macro-F1 = 0.21, $p = 0.015$), and arousal did not reach significance (macro-F1 = 0.36, $p = 0.055$). A partial-correlation analysis ruled out ambient CO₂ as a confounder of the plant–valence relationship. The results indicate a weak but statistically robust association between plant bioelectrical activity and human valence, suggesting replication with larger samples and validated ground-truth measures.

Statement of Original Authorship

We hereby declare that we have completed the present work independently and without the use of any aids other than those specified. All passages that have been taken, either literally or in substance, from published or unpublished writings are identified as such. The work has not previously been submitted, in the same or a similar form, in whole or in part, as part of another examination.

Group 1

Cologne, 31.07.2026

Declaration on the Use of Generative AI

We declare that generative AI tools were used during the preparation of this project, in the following areas:

- **Coding:** Assisting with writing, debugging, and refactoring code across the recording, labeling, dataset-creation, and machine-learning notebooks (sensor recording, emotion labeling, feature engineering, and the Random Forest / XGBoost / Neural Network classification pipelines for valence, arousal, and emotion), the CO₂ correlation analysis, and the live dashboard.
- **Writing:** Drafting sections of this report and supporting documentation, including explanatory text and descriptions of methods and results.
- **Rewriting/editing:** Revising, restructuring, and improving the clarity, grammar, and conciseness of previously written text and code produced by the team.

All analytical decisions, model design choices, interpretation of results, and conclusions presented in this report reflect the judgment of the team members. Generative AI outputs were reviewed, verified, and edited by the team before inclusion; the team takes full responsibility for the accuracy and integrity of the final content.

Table of contents

1	Introduction	1
2	Related Work	1
2.1	Human Emotion Classification based on Plant Bioelectrical Signals	1
2.2	The Circumplex Model	2
2.3	Automated Emotion Recognition Using HSEmotion	3
2.4	Heart Rate Variability as Emotional Scale	3
2.5	Research Gap	3
3	Research Hypotheses	4
3.1	Primary Hypothesis: Plant Signals and Human Emotions	4
3.2	Secondary Hypothesis: Ambient CO ₂ as a Confounding Variable . .	4
4	Method	4
4.1	Architecture Overview	5
4.2	Sensor Setup	6
4.3	Field Work	7
4.4	Labeling Algorithm	8
4.5	Feature Engineering	9
4.6	Machine Learning Models	10
4.7	Live Demonstration	11
4.8	Correlation Analysis	11
5	Results	11
5.1	Model Performance	11
5.1.1	Emotion Classification	12
5.1.2	Arousal Classification	13
5.1.3	Valence Classification	13
5.2	Correlation Analysis	15
6	Discussion	16
6.1	Interpretation of our Findings	16
6.2	Comparison with Previous Studies	18
7	Weaknesses and Future Work	19
7.1	Weaknesses	19
7.2	Future Work	20
8	Conclusion	21
A	Appendix	23
A.1	Plant Feature Columns	23
A.2	Labeling Equations	25
A.3	Example Session Record	27
A.4	Experimental Setup and Live Dashboard	28
A.5	Confusion Matrices	29
A.6	Permutation Tests	31
A.7	Feature Importances	33
A.8	CO ₂ Confound Analysis	35
A.9	Code Repository	36
	References	37

List of Abbreviations

ADC	Analog-to-digital converter
AI	Artificial intelligence
bpm	Beats per minute
CAM	Crassulacean acid metabolism
CNN	Convolutional neural network
CO ₂	Carbon dioxide
COIN	Collaborative Innovation Networks
DC	Direct current
ECG	Electrocardiography
FER	Facial Emotion Recognition
HR	Heart rate
HRV	Heart Rate Variability
IAPS	International Affective Picture System
MAD	Median absolute deviation
ppm	Parts per million
RMSSD	Root mean square of successive differences
ROC-AUC	Receiver operating characteristic — area under the curve
RRI	R-R interval
SAM	Self-Assessment Manikin
VOC	Volatile organic compound

List of Figures

1	Circumplex model of affect	2
2	System pipeline overview	5
3	Classifier comparison for the target variable emotion.	12
4	Classifier comparison for the target variable arousal.	13
5	Classifier comparison for the target variable valence.	14
6	Example session record	27
7	Physical setup of the experiment	28
8	The live dashboard	28
9	Confusion matrices for the target variable emotion	29
10	Confusion matrices for the target variable arousal	30
11	Confusion matrices for the target variable valence	30
12	Permutation test for the target variable valence	31
13	Permutation test for the target variable arousal	32
14	Permutation test for the target variable emotion	32
15	Feature importances for the target variable emotion	33
16	Feature importances for the target variable arousal	33
17	Feature importances for the target variable valence	34
18	Confounding triangle and partial correlations, raw analysis	35
19	Confounding triangle and partial correlations, session-normalized analysis	35

List of Tables

1	Model performance by target variable	15
2	Plant feature columns	23

1 Introduction

Previous research suggests that a plant’s bioelectrical activity may carry information about a nearby human’s emotional state (Gloor 2025), yet the underlying mechanism remains poorly understood. In particular, it is unclear whether the plant’s electrophysiological voltage signal carries information about the human’s emotional state as captured by physiological and behavioral indicators such as Heart Rate Variability (HRV) and facial expression. Clarifying this would provide a first step toward understanding whether plant bioelectrical activity can serve as a viable, non-invasive channel for detecting human emotional states.

In this paper, we develop a plant emotion detector that takes the plant’s voltage signal as input and, based on this signal alone, infers the emotional state of a human seated in front of the plant. Rather than relying on a single subjective report, the human’s emotional state is operationalized through two independent measures: Facial Emotion Recognition (FER) for valence and Heart Rate Variability (HRV) for arousal. A dedicated labeling algorithm, developed as part of this work, combines these two channels into a per-window proxy emotion label along the valence and arousal dimensions of the circumplex model. This leads to the following research question:

Can a human’s emotional state, as measured through FER and HRV, be predicted from electrophysiological plant signals alone, and does any such association persist once ambient CO₂ concentration is controlled for?

To address this question, we conducted an experiment with 15 participants and used the resulting data to train and compare three machine learning models capable of predicting a human’s emotional state exclusively from the plant’s voltage signal.

2 Related Work

This paper investigates whether electrophysiological plant signals carry information about human emotional states, how those states are labeled by Facial Emotion Recognition (FER) and Heart Rate Variability (HRV), and whether ambient CO₂ concentration, recorded alongside, could account for any such relationship. This chapter reviews the relevant background in four areas: plant bioelectricity and human–plant interaction, the theoretical structure of human emotions, automated emotion recognition via FER, and HRV as a scale for emotional arousal. It closes by identifying the resulting research gap.

2.1 Human Emotion Classification based on Plant Bioelectrical Signals

The most important foundation for this study is the pilot investigation about human emotion classification based on plant bioelectrical signals (Gloor 2025). Gloor examines whether bioelectrical signals from a single plant could be used to classify environmental conditions (light on/off) and human emotions. Using an AD8232 ECG sensor, mel-spectrograms, and a ResNet18 CNN, lamp detection was achieved with an accuracy of 85.4%. For emotion classification, it is distinguished between happy and sad based on automated FER labels, resulting in an accuracy of about

73% at the optimal time lag. Different time delays between emotional expression and the plant signal were tested, with the optimal delay identified as one second. For control experiments, Gloor (2025) recorded inside a closed vehicle acting as a Faraday cage and attached the electrodes to aluminum foil in place of the plant (a sham control); these reduce, but do not completely eliminate, the possibility of trivial electromagnetic or equipment artifacts. The study explicitly emphasizes the preliminary, exploratory nature of the findings and the need for further replication (Gloor 2025).

2.2 The Circumplex Model

Both the categorization of emotions into positive and negative valence classes used by Gloor (2025) and the IAPS/SAM methodology used by Choi et al. (2017) are based on the influential circumplex model presented by Russell (1980). Using four methodologically independent scaling procedures, Russell (1980) shows that affective states can consistently be represented as a two-dimensional, bipolar space along the axes of valence (pleasure-displeasure) and arousal (arousal-sleepiness).

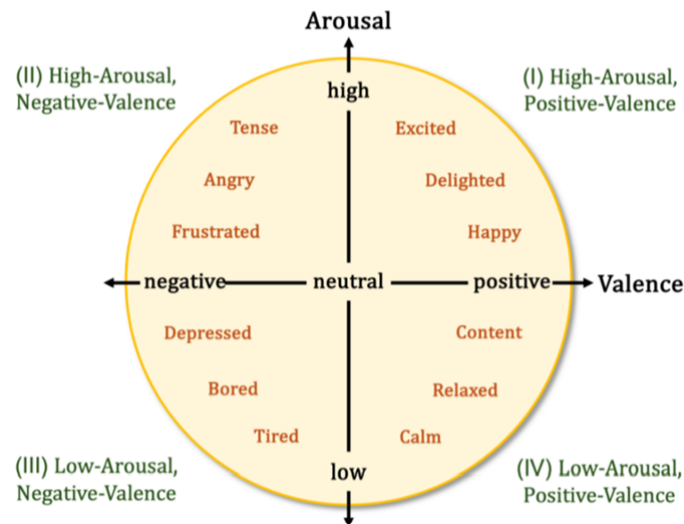


Fig. 1. Circumplex model of affect: the two-dimensional structure of affect, spanned by a valence and an arousal axis, following Russell (1980). Emotion words within one quadrant agree in the sign of both axes. Figure reproduced from Lee et al. (2025).

Figure 1 shows this space. The two axes divide it into four quadrants, each collecting emotion words that agree in the sign of both valence and arousal: *excited*, *delighted*, and *happy* differ in intensity but all combine positive valence with high arousal, and the same holds for the remaining three quadrants. It is this quadrant, rather than the individual emotion word, that the present study treats as the classification target. A recorded window therefore counts as emotionally classified when the classifier places it in one of the four quadrants: high-arousal positive (I), high-arousal

negative (II), low-arousal negative (III), or low-arousal positive (IV), referred to below as *positive activated*, *negative activated*, *negative calm*, and *positive calm*. Windows that stay near the neutral centre of either axis carry no directional information and are assigned the neutral class instead. Section 4.4 describes the rule that implements this crossing.

2.3 Automated Emotion Recognition Using HSEmotion

Savchenko (2022) presents HSEmotion, an open-source library for FER. It is based on EfficientNet architectures and is optimized for speed and mobile use. The models were first trained on a face identification task (VGGFace2 dataset) and then fine-tuned on images from AffectNet to classify emotions. HSEmotion achieved accuracies ranging from approximately 55-70% and inference times of 59-191 ms on mobile devices. These figures illustrate that even highly developed FER systems perform significantly below perfect classification accuracy and may contain systematic, dataset-dependent bias. This is of central importance for the evaluation of automatically generated emotion labels, such as those used in this study, since false classifications of the FER component directly influence the validity of every subsequent analysis that relates those labels to physiological or bioelectrical signals.

2.4 Heart Rate Variability as Emotional Scale

HRV is frequently used to supplement or validate subjective measures of emotion. Choi et al. (2017) evaluated the validity of HRV to label human emotions with 30 participants. Five photos were selected for each of the categories of “happy,” “unhappy,” and “neutral” from among the images provided by the International Affective Picture System (IAPS). The extracted R-R interval (RRI) value is compared to the valence, arousal, and dominance value from the self-assessment. The results show a significant correlation between RRI and the subjective ratings of valence and dominance only for the category of unpleasant and highly arousing stimuli. In contrast, no significant correlations are found for neutral and pleasant stimuli. The authors conclude that HRV can serve as a valid measure of emotion, but only when emotional arousal is high enough. This finding is relevant to the research question of this study, because it suggests that the correlation between plant signals and HRV data may also be modulated by arousal and is not expected to be linear across all emotion categories.

2.5 Research Gap

Taken together, the reviewed work shows that plant bioelectrical signals can be classified into coarse human-emotion categories (Gloor 2025), yet this evidence stems from a single-subject, exploratory design without a standardized elicitation protocol. The circumplex model (Russell 1980) and the documented validity limits of FER (Savchenko 2022) and HRV (Choi et al. 2017) further suggest that a binary happy/sad framing is too coarse to reveal where plant signals actually carry emotional information. It therefore remains open whether the association replicates across multiple participants under a controlled stimulus, and whether it holds separately for valence and arousal. The present study addresses this gap.

3 Research Hypotheses

Building on the theoretical foundations above, we derive two hypotheses from the research gap identified in the related work. The first (H1) states the study’s primary claim, that plant signals carry information about human emotion. The second (H2) is subordinate to it and puts ambient CO₂ forward as a competing explanation for any such information. Throughout, emotion is operationalized according to the circumplex model (Russell 1980) along two dimensions: valence, measured via FER using HSEmotion (Savchenko 2022), and arousal, measured via HRV using a Polar H10 sensor.

3.1 Primary Hypothesis: Plant Signals and Human Emotions

The pilot study shows that bioelectrical plant signals could classify “happy” and “sad” with 73% accuracy using ResNet18 and mel-spectrograms, providing initial empirical evidence of an association (Gloor 2025). Building on this, we test whether the plant signal on its own is sufficient to recover the operationalized emotional state.

Based on this reasoning, we formulate the following hypothesis:

***H1:** A person’s emotional state can be predicted from their electrophysiological plant signals alone.*

3.2 Secondary Hypothesis: Ambient CO₂ as a Confounding Variable

The second hypothesis concerns a competing explanation for any association H1 may yield. The plant is exposed not only to the person but also to the air in the room they breathe in, and ambient CO₂ is related to both sides of the comparison H1 draws. It rises and falls with the participant’s respiration, which itself changes with emotional state, and it plausibly affects the voltage measured between plant and electrode directly. CO₂ therefore occupies the position of a classical confounder, a variable associated with both the predictor and the outcome, and any association observed under H1 could be an artifact of that shared dependence rather than evidence that the plant tracks the person.

Therefore, we formulate the following hypothesis:

***H2:** The association between plant bioelectrical activity and human emotional state is confounded by ambient CO₂ concentration.*

The two hypotheses are tested with different instruments and in a fixed order. H1 is evaluated by supervised classification of the proxy labels from the plant features alone, scored against a permutation null. H2 is then evaluated by partial correlation, restricted to the emotional dimension for which H1 yields an association, since an axis without an effect offers nothing for a confounder to explain.

4 Method

This chapter translates the two hypotheses into a study design and describes how that design was realized. Testing H1 requires three things: an experimental setting in

which the plant and the person are recorded simultaneously, an operational definition of the person's emotional state, and a classifier that is given the plant signal and nothing else. Testing H2 requires that ambient CO_2 be recorded during the same sessions, so that the competing explanation can be examined on the study's data.

Three commitments follow from this and hold for the rest of the chapter. First, emotion is measured rather than asked for, on two independent channels: FER for valence and HRV for arousal, crossed into one proxy label per window. Second, these two channels serve only to construct that label. The plant voltage is the sole predictor the models ever receive, and the separation is maintained through feature extraction, training, and evaluation, so that the result cannot be an artifact of human-derived features entering the input. Third, CO_2 is recorded throughout and reserved for the analysis that tests H2.

The chapter follows the data along this path: an overview of the pipeline, the sensor hardware, the recording sessions, the labeling algorithm that produces the proxy labels, the features extracted from the plant signal, the models and their evaluation, the live dashboard, which demonstrates the result in real time, and finally the correlation analysis that closes the argument by testing H2.

4.1 Architecture Overview

The system is organized as a sequential data pipeline that transforms raw multimodal sensor recordings into a trained plant-based emotion classifier, complemented by a separate real-time inference component, a live dashboard. Figure 2 provides an overview. The pipeline consists of five stages, described in turn below: sensing, preprocessing, labeling, modeling, and live application.

Sensing. Four independent sensors capture the raw data: plant electrodes record

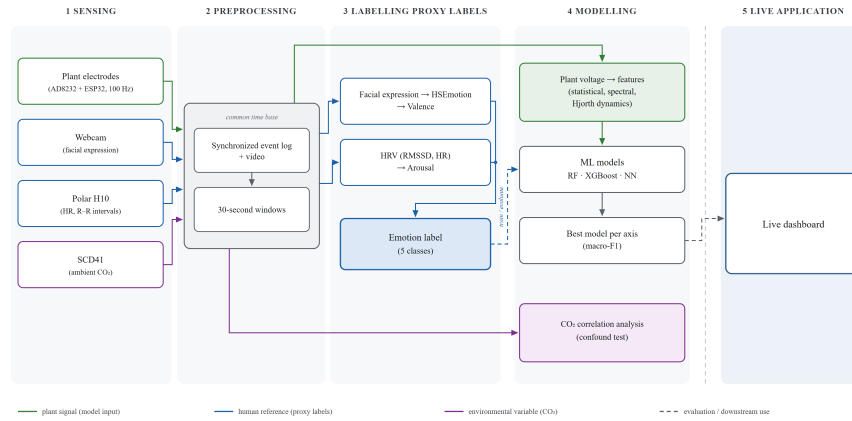


Fig. 2. System pipeline overview: colour marks the role each stream plays: green is the plant signal, the only family the classifiers ever see; blue is the human reference from which the proxy labels are derived; purple is the environmental variable used for the confound test. Dashed arrows denote evaluation and downstream use rather than data flow.

the plant bioelectric voltage, a webcam records facial expressions, a Polar H10 chest strap records heart rate and R-R intervals, and an SCD41 sensor records ambient CO₂. The sensor setup builds on the implementation by Gloor (2025).

Preprocessing. Each session consists of one participant watching a sequence of short emotion-eliciting video clips while the four streams are recorded. The sensors run independently; their readings are aligned into one shared event log only after recording, together with the corresponding video file. This common time base is then divided into non-overlapping 30-second windows, which form the unit of analysis for every stage that follows.

Labeling. The windows are turned into a proxy emotion label. Valence is derived from facial expression via HSEmotion and arousal from HRV; the two are crossed into a five-class emotion label.

Modeling. The plant voltage of each window is reduced to a set of statistical, spectral, and signal-dynamics features, and the labeled windows of all sessions are merged into a single consolidated dataset. Three classifiers, namely Random Forest, XGBoost, and Neural Network, are then trained and compared separately for emotion, arousal, and valence, using plant-bioelectric features only, and the best model per axis is selected by macro-averaged F1.

Live application. The selected models are reused in the live dashboard, which displays the sensor and model outputs and infers emotion in real time.

Alongside these stages, the CO₂ correlation analysis tests whether ambient CO₂, rather than the person's emotional state, could account for whatever association the classifiers establish.

The complete implementation of every stage described in this chapter is publicly available in the project repository. This covers the firmware, recording, labeling, dataset creation, the three model pipelines, the correlation analysis, and the live dashboard. Section A.9 gives the address and maps each stage to the file that implements it.

4.2 Sensor Setup

The sensing stage relies on four independent sensors: plant electrodes, a camera, the Polar H10, and the SCD41 CO₂ sensor. This section describes the hardware behind each of them and the program that records them, starting with the plant sensing setup itself.

The plant sensing hardware builds on the ESP32 and AD8232 architecture previously used for plant biosensing by Gloor (2025). The AD8232, originally designed as a heart-rate front end, is repurposed here to amplify the weak bioelectric signal of the plant. The ESP32 microcontroller samples the amplified signal at 380 Hz internally, decimates it, and streams the result over WebSocket at a configured output rate of 100 Hz. Because the decimation factor is an integer, the rate actually delivered is higher than the configured one: measured against the recording timestamps it is approximately 119 samples per second. Every rate-dependent feature is therefore computed against the rate measured for that session rather than the nominal one, and the pipeline refuses to window a session whose measured rate departs from the expected value by more than ten percent. Only this hardware front end was

adopted; the recording, synchronization, labeling, and machine-learning pipeline were developed independently for this project.

In addition to the plant voltage, the ESP32 is also connected to an SCD41 sensor, which reports ambient CO₂ concentration in ppm. The SCD41 measures on its own fixed cycle of approximately five seconds, so a 30-second window rests on about six independent readings.

Two further sensors complete the setup: a Polar H10 chest strap, connected via Bluetooth, providing R-R intervals for the later HRV-based arousal label, and the laptop's integrated webcam recording the participant's face for the FER-based valence label.

A single *Kalanchoe* plant was used throughout all recording sessions. Three standard medical electrodes used for ECG recordings were attached to the three largest leaves of the plant. Each electrode was connected to the AD8232 board, which forwarded the resulting voltage signal to the ESP32 for transmission.

A recording program was developed to drive each session. It opens the three acquisition paths, the WebSocket connection to the ESP32, the Bluetooth connection to the Polar H10, and the webcam capture, and runs them concurrently for the duration of the session. None of the four sensors acts as a master clock; instead, each sensor writes its readings independently into one shared event log, timestamping every entry against the local system clock of the recording laptop. The output of a completed session is this shared event log together with the corresponding video recording, which together form the raw input to every following processing step.

The entire sensor setup was assembled and operated in the Digital Lab of the Cologne Institute for Information Systems.

4.3 Field Work

Participants were recruited from fellow students, personal acquaintances, and other students unknown to the team beforehand. All participants were between 20 and 30 years old and from Germany. In total, 15 participants completed one session each.

The emotional stimulus consisted of a video of approximately ten minutes in length, reused from an earlier COIN seminar project (Schäfer 2025). The video follows a fixed sequence of emotionally contrasting segments, beginning with light-hearted content such as clips of dogs and a humorous advertisement, and later introducing material intended to evoke disgust, shock, and sadness, including scenes of a more graphic and violent nature. It ends with an extended unsettling sequence that concludes with a jump scare. All participants viewed the clips in the same fixed order.

During the session, participants were seated at a table facing the laptop and asked to watch the video without any additional task. Participants were informed verbally about the nature of the video, including the possibility of disturbing or graphic content, and were asked whether there were any topics they did not want to be confronted with. Each participant completed the session alone in the room, and the room was ventilated between sessions.

Each session opens with a 30-second resting baseline, before the video starts, during which the participant is asked to sit still. This ensures that the session starts

from a resting state. Plant voltage and R-R intervals are recorded from the start of the session, including this baseline, while the camera stays open but writes no frames until the baseline has ended. The windows covering the baseline therefore carry no valence value and are excluded rather than labeled.

The participant sat at the table directly in front of the laptop running the video and the recording code. The plant with the attached electrodes stood on the table to the left of the laptop, well within the participant’s field of view, and the breadboard carrying the AD8232, the SCD41 and the ESP32 sat in front of the plant, connected to it by the electrode leads. Figure 7 in Section A.4 documents this arrangement.

4.4 Labeling Algorithm

Self-assessment ratings, for instance following the Self-Assessment Manikin (Bradley and Lang 1994), would provide a direct subjective report of the emotional state for each window. They were deliberately not used here, because such ratings may be influenced by conscious or unconscious response bias and therefore do not serve as an independent ground truth. Instead, valence and arousal are derived from separate physiological and behavioral channels, processed independently, and combined in a final step to form the crossed emotion label.

Before these channels can be processed, the independently timestamped sensor streams must first be brought onto a common time grid. Each session is divided into 30-second windows based on elapsed recording time, and every sensor reading is assigned to the window whose interval contains its timestamp; only once the plant voltage, facial video, and R-R interval series have been aligned in this way can values from different sensors within the same window be treated as referring to the same moment in the session. This window length follows Baek et al. (2015), who show that 30 seconds is the shortest interval from which HRV metrics can still reliably approximate the much longer five-minute windows recommended by standard HRV guidelines. These 30-second windows are non-overlapping and form the grid used for calibration and model training.

Before any HRV measure is computed, the raw R-R interval series recorded by the Polar H10 is corrected for artifacts using the beat classification method of Lipponen and Tarvainen (2019), implemented in NeuroKit2 (Makowski et al. 2021). A single misdetected heartbeat can otherwise distort every downstream HRV value, so correction is applied per window before the arousal signal is derived.

Valence is derived from facial expressions. Video frames are sampled at 5 Hz and each sampled frame is passed through HSEmotion (Savchenko 2022), a model trained on the AffectNet corpus (Mollahosseini et al. 2019) that outputs a continuous valence estimate per detected face. These per-frame estimates are averaged per window, weighted by the model’s own confidence, to obtain a single valence value for that window (Equation 1). Windows in which a face was detected in too small a share of the sampled frames are excluded rather than labeled (Equation 2).

Arousal is derived from HRV by combining RMSSD with mean heart rate into a single index. Both quantities are computed per window from the corrected R-R series (Equation 3, Equation 4). Since RMSSD decreases with physiological activation while heart rate increases with it (Kreibitz 2010), RMSSD is inverted before the two

measures are averaged. Both quantities enter this index as robust z-scores relative to the session’s own median and spread (Equation 5), yielding the single arousal index of Equation 6.

Because resting heart rate, HRV, and facial expression differ substantially between individuals, no fixed threshold could separate an emotionally salient window from a neutral one across all participants: a resting heart rate of 90 bpm might indicate heightened activation for one person and be entirely unremarkable for another. Each session’s own data therefore establishes what counts as that participant’s baseline on that day, and every threshold below is expressed relative to it.

Both axes are discretized by the same rule, applied independently. A window is classified as positive or negative in valence, or as activated or calm in arousal, when its value lies more than a deadband half-width τ from that session’s median on that axis; otherwise it falls into the respective neutral category (Equation 8, Equation 9). Median and MAD are recomputed for each session from its own non-overlapping 30-second windows, and τ follows from that spread (Equation 7). Only the multiplier k in Equation 7 is fixed, at 0.5 for both axes, setting the deadband half-width to about half a robust standard deviation. It was chosen once before the main analysis by inspecting a pilot session, so that a substantial share of windows still received a directional label rather than collapsing into the neutral category.

The two axis labels are then crossed into the final emotion label (Equation 10). A window is labeled *neutral* whenever either axis is neutral; only when both are directional does it receive one of the four combinations, *negative activated*, *negative calm*, *positive activated*, or *positive calm*. This follows the quadrant structure of Russell’s circumplex model (Russell 1980), shown in Figure 1.

Figure 6 in Section A.3 shows the procedure applied to one complete recording, from the two input channels through the session medians to the crossed label of every window.

4.5 Feature Engineering

For every 30-second window, a set of features is extracted from the plant voltage signal alone. These fall into three families: (i) time-domain amplitude and shape descriptors (mean, median, standard deviation, detrended standard deviation, interquartile range, range, root-mean-square, zero-crossing rate); (ii) spectral descriptors from a Welch power spectrum (band powers in four frequency bands, their ratios, and spectral entropy); and (iii) signal-dynamics descriptors (Hjorth activity, mobility, and complexity, and a spike rate counting transient excursions beyond three robust standard deviations from the median). This yields 22 feature columns per window; Table 2 lists them individually with their definitions, together with the six quality columns that accompany them into the design matrix.

Each feature is provided in two representations that are compared during training: *raw*, the value on its original absolute scale, and *session-normalized*, a per-session robust z-score obtained by subtracting the session median and dividing by the scaled median absolute deviation computed over that session’s own non-overlapping windows (Equation 11). The session-normalized representation removes between-session offsets in the plant signal that are unrelated to the participant’s emotional state.

To assemble the training data, the labeling pipeline is run over every recorded session, producing one file of non-overlapping labeled windows per session (one row per window). The presence of a labeled file is verified for each session; all files are then loaded, checked for an identical column schema, and concatenated into one consolidated dataset.

4.6 Machine Learning Models

The classifier predicts the arousal, valence, and emotion label of each 30-second window from the plant-bioelectric features only. The classes are: arousal (calm, neutral, activated), valence (negative, neutral, positive), and emotion (negative calm, negative activated, neutral, positive calm, positive activated).

Before training, rows without a target label are removed, remaining missing feature values are imputed with the column median, constant (non-informative) features are dropped, and the target is label-encoded. Three classifiers are then trained and compared, each on both feature representations (raw and session-normalized), giving six combinations per target:

- **Random Forest** with balanced class weights;
- **XGBoost**;
- **Neural Network** (multilayer perceptron) wrapped in a StandardScaler pipeline

These three names are used throughout the paper. The comparison figures in Section 5 are generated by the pipeline itself and therefore label the same models with their code identifiers `RandomForest`, `XGBoost`, and `NeuralNet`, and the two representations as `raw` and `session_norm`.

Because the emotion target is strongly imbalanced, with a large neutral class dominating (62.3%), overall accuracy would largely reflect the majority class. Macro-averaged F1, which weights every class equally, is therefore used as the primary selection and comparison metric, complemented by accuracy, one-vs-rest ROC-AUC, a confusion matrix, and a per-class classification report.

Models are evaluated with a nested, participant-grouped, stratified k-fold cross-validation:

- windows are **grouped by participant**, so that no participant appears in both the training and test portion of a fold, preventing participant-level leakage;
- folds are **stratified** by class to keep class proportions comparable across folds;
- the **outer** loop (5 folds) provides the held-out test estimate, while an **inner** loop (3 folds) performs hyperparameter tuning by grid search on each outer fold's training data only, so the test folds never influence tuning.

For each target, the model with the highest macro-F1 wins and is persisted for the live dashboard. Whether that score could have arisen by chance is then checked with a label-permutation test: the target is shuffled 200 times, and each shuffle is rescored by refitting the winning architecture across the same participant-grouped folds. The null distribution is then compared against the observed score to evaluate the significance of the observed score.

4.7 Live Demonstration

To demonstrate the whole setup, we developed a single dashboard that shows every data source at once. This includes the live signals detected by the sensors: plant bioelectric voltage, CO₂, heart rate, and camera feed. In addition, two values per axis (arousal, valence, emotion) are calculated and displayed side by side. On the right side, it shows the proxy labels, applied live to HRV and FER using the same labeling rule. On the left side, it presents the corresponding plant-model predictions of arousal, valence, and emotion. These predictions originate from the pre-trained models; for each axis, the model with the best macro-F1 score is applied to every 30-second window.

Placing the two values per axis directly next to each other makes the comparison between the plant model and the labeling algorithm the visual centre of the display. Figure 8 in Section A.4 shows the dashboard during a session, captured in a window in which the two agree on positive valence but disagree on arousal, which is the qualitative pattern that the quantitative results in Section 5 reproduce.

Every live signal updates continuously. Only the two predictions per axis are window-based. Both recompute once every window of data has been collected. There is no training here; the winning models are only loaded once. To include the participant's individual variability, the session normalization calibration is recomputed after each completed window in the live demonstration. Therefore, it is growing across the live session and updates in real time. After each new window, the median and MAD are recalculated based on all previous windows. At the beginning of a session, the classification is therefore less reliable than later on.

4.8 Correlation Analysis

The correlation analysis is the study's secondary test and is used to examine H2.

We test the three pairwise relationship branches a CO₂ confound would require (plant-label, plant-CO₂, CO₂-label) and use the partial Spearman correlation $\rho(\text{plant, label} \mid \text{CO}_2)$ as the decisive test. Both are computed across all 22 plant features (Table 2) in two complementary variants: a raw analysis and a session-normalized analysis. A branch counts as present when it reaches $|\rho| = 0.15$ at $p < 0.05$, and on the two branches measured per feature at least 40 % of the features must clear that bar. The threshold for the amount of features is arbitrary and was chosen so that a significant amount of plant features actually needs to correlate with CO₂ for the branch to be considered present. Every correlation relates values of the same 30-second window, so the branches and the partial correlation all refer to the same CO₂ values. If the plant-label association collapses after controlling for the partial correlation, CO₂ is assumed to account for it; if it survives, H2 is rejected.

5 Results

5.1 Model Performance

To answer the research question of whether the bioelectrical plant signal predicts the participant's emotional state, three machine learning model architectures are

compared (Random Forest, XGBoost, Neural Network), each of them tested on two feature representations (raw, session-normalized), resulting in six combinations for each of the three target variables: valence, arousal, and the proxy emotion label.

Data recording resulted in a dataset including 284 windows from 15 participants. The evaluation was performed using a nested five-fold cross-validation procedure grouped by participants, ensuring that no participant appeared in both the training and test data simultaneously and that the hyperparameter tuning never utilized information from the test data used to evaluate how well the model performs. This ensures that the reported values are reliable and avoids leakage. For statistical validation, a permutation test with 200 label permutations was additionally performed to assess how likely the obtained result would be under the null hypothesis that the plant features carry no information about the label.

5.1.1 Emotion Classification

Regarding our research question, our most important target variable is emotion. Similar to the Circumplex Model (Russell 1980), our emotion label is created by combining the valence and arousal measurements, leading to five classes: negative activated, negative calm, neutral, positive activated, positive calm. The best-performing combination is Random Forest on the raw representation, with an accuracy of 42% and a macro-F1 score of 0.21 (see Figure 3).

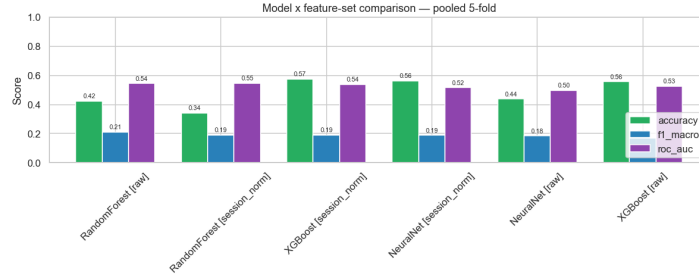


Fig. 3. Classifier comparison for the target variable emotion.

This is the only case where the raw representation performs better than the session-normalized one, contradicting the otherwise consistent trend. Regarding the target variable emotion, there is a significant class imbalance, as “neutral” dominates: 177 of the 284 windows carry it, and it is the only class the winning model recognizes reliably (F1 0.64). The model largely predicts the majority class well while barely distinguishing the minority classes (F1 0.08 for “positive activated”, 0.07 for “negative calm”), as reflected in the much lower macro-F1 of 0.21 (Figure 9). So while emotion prediction is statistically better than a random classifier, accuracy alone overstates how much the model actually understands about emotional states. The permutation test verified the statistical significance ($p = 0.015$; Figure 14), but it is weak as two permutations were still able to beat the observed score. The feature ranking for this target (Figure 15) is led by `plant_max`, `ratio_bp1_5_bp5_15`, and `plant_median`, that is by amplitude and spectral-shape descriptors, with `spectral_entropy` and the

Hjorth dynamics descriptors following. This is a different profile from the other two targets, and one to be read with the caveat that this model largely reproduces the majority class.

5.1.2 Arousal Classification

The target variable arousal, derived from HRV (RMSSD) and the mean heart rate, contains three classes: calm, neutral, and activated. None of the classifiers are able to predict arousal better than random: across the six combinations, accuracy ranges from 32% to 37%, which is close to the approximately 33% expected by chance with three classes (see Figure 4).

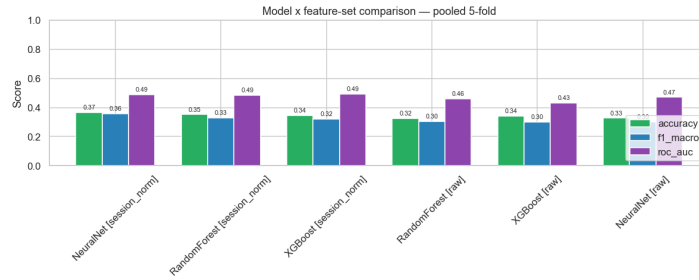


Fig. 4. Classifier comparison for the target variable arousal.

The Neural Network on the session-normalized features performed the best, at an accuracy of 37% and a macro-F1 of 0.36. The result does not reach the threshold to be statistically significant ($p = 0.055$; 10 of 200 permutations matched or exceeded it; Figure 13). The classification report shows that this combination classified all three classes with similarly low accuracy, at per-class F1 scores of 0.35 (activated), 0.30 (calm), and 0.42 (neutral), so no class is recognized appreciably better than the others (Figure 10). However, it must be noted that there is an unbalanced distribution of the classes: “neutral” is clearly the largest class, with 115 out of 284 cases (approximately 40.5%), while 80 cases are labeled activated, and 89 calm. The approach of session normalization is most effective and consistent for the target variable arousal, improving the macro-F1 scores of all three models, by an average of 0.035. As with valence, DC-invariant and session-relative features dominate the ranking. Because the winning neural network exposes no feature importances, the ranking is read from the tree-based models on the same session-normalized representation (Figure 16); it is led by `spike_rate` (the frequency of transient amplitude fluctuations), `plant_max`, and `bp_15_30`. XGBoost orders its own ranking differently, headed by `hjorth_activity` and `plant_detrend_std`, but it too places all three of those features in its top five, so the trio is not an artifact of a single model.

5.1.3 Valence Classification

The target variable valence, determined by FER (HSEmotion), contains three classes: negative, neutral, and positive. Across the six combinations, accuracy ranges from

36% to 43%. Figure 5 represents the macro-F1 score, the accuracy and the ROC-AUC of all six combinations sorted by their macro-F1 value.

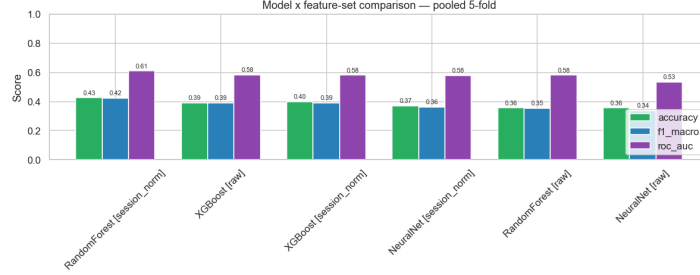


Fig. 5. Classifier comparison for the target variable valence.

Of the six combinations compared, Random Forest on the session-normalized features performed the best, with an accuracy of 43% and a macro-F1 of 0.42. The permutation test confirms that this result is statistically significant ($p = 0.005$; Figure 12). Zero out of 200 random label permutations matched or exceeded the result actually observed. Therefore, valence is a target variable for which the plant signal shows a clear and robustly validated predictability. In addition, all three classes of valence were recognized with similar accuracy: the winning model predicted negative, neutral, and positive with per-class F1 scores of 0.43, 0.45, and 0.40 respectively (Figure 11). In contrast to the emotion classification, there was no significant bias toward a dominant class in this case. Averaged across all three models, the effect of session normalization increases the macro-F1 scores by 0.031 for this target variable.

The winning model's two highest-ranked predictors are the Hjorth dynamics descriptors `hjorth_mobility` (how quickly the signal changes) and `hjorth_complexity` (how that rate of change itself varies), followed by `plant_min`, `ratio_bp1_5_bp5_15`, and `plant_range` (Figure 17). XGBoost, tuned independently on the same representation, puts the same two Hjorth descriptors at the top, so the ranking does not depend on which tree model is asked. The dynamics descriptors are level-invariant by construction, and the amplitude descriptors behind them enter only in their session-normalized form, so they express how far a window departs from that session's own baseline rather than where the session sits in absolute terms. The model therefore relies mainly on how the curve moves within the window, not on where it lies overall. This is an indication that the model is learning about the actual signal dynamics and not recognizing a session based on its random baseline.

Table 1 summarizes the best combination per target variable.

Table 1. Model performance by target variable: best (model, representation) combination per target, with its pooled nested cross-validation scores and its permutation-test p-value.

Target	Best model	Accuracy	Macro-F1	p
Valence	Random Forest (session-normalized)	43%	0.42	0.005
Arousal	Neural Network (session-normalized)	37%	0.36	0.055
Emotion	Random Forest (raw)	42%	0.21	0.015

5.2 Correlation Analysis

H2 is assessed solely on valence, as it is the only label for which the plant features demonstrated a statistically meaningful association and actual predictive performance.

The correlation analysis is conducted in two complementary variants, one per figure in Section A.8. In the raw analysis (Figure 18), plant features showed a weak but frequently significant correlation with valence (mean $|\rho| = 0.14$, significant in 13 of 22 features), while the correlation between plant features and CO₂ concentration was substantially stronger (mean $|\rho| = 0.40$, significant in 18 of 22 features). CO₂ itself was also correlated with valence ($\rho = 0.31$, $p < 0.0001$). Together, these three pairwise relationships form a plausible confounding triangle: CO₂ correlates with both the plant signal and valence, raising the question of whether the plant–valence relationship is genuine or merely an artifact of CO₂ acting on both variables independently. To test this directly, we computed the partial correlation between plant features and valence while controlling for CO₂. If CO₂ were the true driver of the plant–valence association, this partial correlation should collapse toward zero. Instead it barely moved: the mean $|\rho|$ went from 0.14 to 0.15, a minor change and in the direction opposite to the one a confound would produce, with the same 13 of 22 features remaining significant. Controlling for CO₂ therefore removes none of the plant–valence association.

The session-normalized analysis (Figure 19) provided further insight into the nature of the CO₂ effect. It repeats all three edges of the triangle on the same 22 features after every signal, the plant features as well as CO₂ and valence, has been centred on its own session, so that only within-session variation remains. Once between-session variance (e.g., differences in room climate across recording days) was removed, the plant–CO₂ correlation dropped sharply to a mean $|\rho|$ of 0.067, significant in only 3 of 22 features, and was no longer classified as a robust, present effect. This indicates that most of the raw plant–CO₂ coupling reflects slow, session-level environmental drift rather than a genuine within-session physiological interaction. The CO₂–valence edge behaved differently and did not weaken ($\rho = 0.33$, $p < 0.0001$), so it is a within-session relationship rather than a between-session one. The plant–valence correlation itself is much weaker within sessions than across

them (mean $|\rho| = 0.07$, significant in 5 of 22 features). That drop is expected rather than contradictory: roughly half of the raw correlation lives in the between-session differences, which is precisely the variance the participant-grouped cross-validation of Section 5 keeps the classifier from using. Despite the reduction of the CO₂ edge, the plant–valence partial correlation remained stable under CO₂ control within sessions as well (mean $|\rho|$ 0.07 before and after control, 5 of 22 significant features before and 6 of 22 after), confirming the confound-rejection result at both the raw and session-normalized level.

6 Discussion

Returning to the hypotheses, the results provide partial support for H1: the plant signal predicts valence above chance (macro-F1 = 0.42, $p = 0.005$) and, although very weakly, the crossed emotion label (macro-F1 = 0.21, $p = 0.015$), whereas arousal does not reach conventional significance (macro-F1 = 0.36, $p = 0.055$). H1 is therefore supported for valence and for the crossed emotion label, but not for arousal.

The secondary analysis addresses a plausible alternative explanation for that result. H2 is rejected, tested on valence as the dimension for which H1 shows the strongest evidence. Both branches of the confounding pathway are in fact present. CO₂ correlates with valence ($\rho = 0.308$, $p < 0.0001$) and, in the raw analysis, with most plant features, so the alternative explanation was not implausible a priori. The decisive condition nonetheless fails: the plant–valence association remains essentially unchanged once CO₂ is controlled for, so it is not an artifact of a shared dependence on ambient CO₂.

6.1 Interpretation of our Findings

Valence is the target on which the positive claim of this study rests. The winning model recognizes all three valence classes at comparable rates, with per-class F1 scores of 0.43, 0.45, and 0.40 (Figure 11), so its macro-F1 of 0.42 does not arise from following the largest class but from a discrimination spread across the entire label space, and none of the 200 label permutations matched the observed score (Figure 12). The effect is nonetheless small in absolute terms: 43% accuracy against the approximately 33% expected by chance with three classes. What the plant signal supports is therefore a statistical association that becomes visible across 284 windows and 15 participants, not a channel from which the emotional state of an individual person could be read off at a practically useful rate.

The feature ranking of that model indicates which properties of the signal carry the association. The two highest-ranked predictors are Hjorth mobility and complexity, dimensionless descriptors of how quickly the voltage changes and how variable that rate of change itself is, and Random Forest and XGBoost place the same pair at the top independently of one another (Figure 17). The amplitude descriptors ranked behind them, `plant_min` and `plant_range`, enter only in their session-normalized form and therefore express how far a window departs from the baseline of its own session rather than where that session lies on the absolute voltage scale. The comparison of the two feature representations is consistent with this. Session

normalization raised the average macro-F1 across the three classifiers by 0.031 for valence, with the Random Forest gaining 0.068, and the best combination for this target used session-normalized features. Since every recording session contains exactly one participant, the absolute voltage level contained in the raw session data partly encodes the session itself, carrying the ambient conditions of the recording and the state of the electrode contact. That performance improves once this level is removed indicates that the association is carried by the dynamics of the signal rather than by properties that distinguish one recording from another.

The crossed emotion label behaves differently. Its macro-F1 of 0.21 lies close to the 0.20 expected from uniform guessing across five classes, and the winning model attains its 42% accuracy almost entirely on the dominant neutral class, with an F1 of 0.64 there against 0.07 and 0.08 for the two smallest classes (Figure 9). Two effects work against this target at once. First, crossing the axes distributes 284 windows over five cells, leaving as few as 21 windows in the smallest of them, so the minority classes are rare. Second, one of the two constituent axes carries no usable signal: a label formed by crossing valence with arousal cannot be more informative than its weaker component, and arousal does not separate above chance. The crossed label therefore does not aggregate the two dimensions; it combines an axis that separates above chance with one that does not. The near-chance emotion result is thus consistent with the valence result rather than in tension with it, and supports treating valence and arousal as analytically distinct targets rather than collapsing them into a single categorical label.

Emotion is also the only target whose best combination uses raw features, which does not reverse the general trend. The other two classifiers improve under session normalization on this target as well, XGBoost from 0.17 to 0.19 and the Neural Network from 0.18 to 0.19, so the reversal is confined to the Random Forest, whose raw advantage rests on the neutral class alone (F1 0.64 raw against 0.57 session-normalized) rather than on any better separation of the minority classes. Given that all six combinations fall between macro-F1 0.17 and 0.21, this difference is within the variation among near-equivalent models.

Arousal is the weakest target, missing conventional significance at $p = 0.055$ with 10 of 200 permutations matching or exceeding the observed score (Figure 13). This null result is difficult to attribute to the plant, because valence and arousal were operationalized through proxies of unequal quality: a facial expression model reading the emotional display itself on the one side, a single index derived from RMSSD and mean heart rate on the other. HRV markers of this kind track emotional state reliably mainly for unpleasant and highly arousing stimuli (Choi et al. 2017), while the video combines strongly negative segments with mild, happy and neutral ones, so part of the recorded material falls outside the range in which the arousal label is expected to be informative at all. A weak label bounds the performance attainable by any predictor of it, including a fully informative one. The arousal result is therefore best read as uninformative about the plant rather than as evidence against it.

The correlation analysis reports much weaker numbers than the classification, and the two can easily be read as contradicting each other. Within sessions, the average absolute correlation between a single plant feature and valence is 0.07, and

only 5 of the 22 features reach significance, yet a classifier trained on those same features reaches a macro-F1 of 0.42. The two quantities measure different things. A Spearman coefficient asks whether one feature on its own rises and falls with valence, whereas the classifier may use all 22 features at once and split each of them at several thresholds, so it can exploit combinations that no individual coefficient registers. The weak correlations therefore do not undercut the classification result. They indicate that the information lies in the combination of the features rather than in any one of them.

Beyond rejecting H2, the CO₂ analysis provides a further result. Separating between-session from within-session variance distinguishes the three edges of the confounding triangle. The plant–CO₂ coupling nearly disappears within sessions (mean $|\rho| = 0.067$, significant in 3 of 22 features), identifying it as slow, room-level drift across the recording day. The CO₂–valence edge, in contrast, persists undiminished within sessions ($\rho = 0.33$, $p < 0.0001$), which could be attributed to respiration changing alongside emotional state on a timescale of minutes. Ambient CO₂ therefore does track the participant, but the plant does not track ambient CO₂ on the timescale at which the classification operates. Ambient CO₂ is thus not a plausible carrier of the valence information that is picked up by the classifier, and the mechanism behind the association remains open.

6.2 Comparison with Previous Studies

The present study directly addresses the two central limitations that Gloor (2025) explicitly identified: the lack of a standardized elicitation protocol and the single-subject design. By employing a fixed, ten-minute video stimulus instead of post-hoc labeling of unconstrained behaviour, and by recording 15 participants with grouped cross-validation to prevent participant-level leakage, the present study constitutes a multi-participant replication of this experimental paradigm with a controlled emotion-elicitation protocol.

The shift from a binary “happy”/“sad” classification to a dimensional valence-arousal framework (Russell 1980) further extends this replication methodologically: it allows for a more fine-grained analysis than Gloor’s binary scheme permitted, and reveals that plant signals are informative for valence but not for arousal, a distinction that could not have been captured within a binary classification framework in the first place.

Both studies validate their findings via permutation testing, so neither rests on cross-validation scores alone. A direct comparison of the reported accuracies is nevertheless of limited value. Gloor (2025) reports 72.8–73.8% at the optimal 1 s lag on a binary valence task (positive: happy/surprise, $n = 968$; negative: fear/anger/sad/disgust, $n = 651$), a 60/40 split. That accuracy sits about 13 points above the majority baseline he reports (59.9%), but his published confusion matrices show the classifier leans heavily on the majority class: sad-class recall is only about 0.52–0.57, giving a macro-F1 of about 0.71, slightly below the headline figure. More important than the accuracy gap, the two evaluation protocols differ in kind: our estimates come from participant-grouped nested cross-validation, whereas Gloor uses stratified k-fold on non-overlapping snippets without grouping by session or day,

so temporally adjacent snippets from the same 30-minute session can fall in both train and test. His label-permutation test does not rule this out, since shuffling labels leaves no session structure for the null model to exploit. The three-class valence result here (36–43%; chance about 33%) therefore cannot be placed on the same scale as his within-subject binary figure, and the comparison supports little beyond the coarse observation that both studies report performance above chance.

7 Weaknesses and Future Work

7.1 Weaknesses

Several limitations qualify the interpretation of the present findings. Most fundamentally, no ground-truth measure of emotional state was collected against which the derived labels could be validated. Valence and arousal rest on two independent proxy measures, facial expression via HSEmotion and HRV via RMSSD, rather than on validated psychological assessments, and because these channels are processed separately and combined only afterwards, any inaccuracy specific to either propagates directly into the resulting label without cross-validation between them.

A structural confound follows from the fixed stimulus order. Since all participants viewed the same video in the same sequence, labels and plant recording share a common temporal profile: the labels track the stimulus timeline, while the plant signal drifts over the session through electrode settling and changing contact impedance. A model can therefore exploit position within the session in the absence of any plant–emotion relationship. Grouping by participant does not remove this, since the temporal structure is shared across groups rather than nested within them, and session normalization removes level but not the shape of the drift.

Construct validity is a related concern. HSEmotion reports categorical accuracies of roughly 55–70% on standard benchmarks (Savchenko 2022); since these refer to discrete categories, confusions within the same valence sign do not propagate into the valence label, so the effective valence error is lower than the categorical figure suggests, though it was not quantified here. The arousal axis rests on a single HRV measure whose validity at lower arousal levels has been questioned (Choi et al. 2017), which may have contributed to the weak arousal result, although a genuinely weak underlying effect is equally consistent with the data. Discretizing both axes adds further researcher degrees of freedom: the thresholds and the rule collapsing them into the five-class label were fixed a priori and not varied.

The stimulus constitutes a further limitation. The video used to elicit emotional responses was reused from an earlier, unrelated COIN seminar project (Schäfer 2025) rather than being normed on valence and arousal in the manner of standardized stimulus sets such as IAPS. Its capacity to reliably elicit the intended range of affective states across a heterogeneous sample was therefore not independently confirmed.

The sample comprises 15 participants and 284 analyzable windows. Since windows are autocorrelated and nested within participants, the effective sample size is closer to the number of participants than of windows, and interval estimates derived at the window level would understate the true uncertainty. The correlation analysis

additionally evaluated 22 features across two analysis variants without correcting for multiple comparisons, which should be taken into account when interpreting the reported significance levels.

As in Gloor (2025), a single plant species (*Kalanchoe*) was used throughout, so generalization across species or specimens remains untested. *Kalanchoe* is moreover a CAM species whose stomata open primarily at night, which limits daytime gas exchange and restricts the CO₂ control analysis in particular, since daytime plant–CO₂ coupling would be expected to be stronger in C₃ species. Unlike Gloor (2025), no sham or shielding controls were included.

The measurement chain imposes its own constraint. The AD8232 is an analog front end designed for electrocardiography, and its passband is optimized for cardiac rather than plant biopotentials, whose action and variation potentials evolve over seconds to minutes. The recorded signal is therefore band-limited, with slow components attenuated by design, which qualifies the feature-importance finding below: the weak contribution of absolute-level features may partly reflect the amplifier rather than the plant.

Finally, all sessions were conducted on a single day in a room not isolated from external influences. Room temperature, humidity, and air movement were neither held constant nor recorded, and their influence on the plant signal cannot be determined from the present data. Only ambient CO₂ was recorded as a mediating channel; participant movement, which generates air displacement and vibration near the plant, cannot be excluded as an alternative pathway.

7.2 Future Work

Several directions follow from these findings. The lower accuracy observed here relative to Gloor (2025) is consistent with the more conservative and generalizable evaluation framework applied, namely grouped cross-validation across 15 participants. This interpretation is directly testable: re-evaluating the present data under a non-grouped, randomly split cross-validation would indicate how much of the difference is attributable to evaluation protocol rather than to effect size. Larger and more diverse samples under professionally controlled laboratory conditions would additionally narrow the wide confidence intervals observed here and minimize session-level environmental drift.

The confounds identified above translate directly into design refinements. Counterbalancing stimulus order across participants decorrelates the emotion timeline from session time, a time-only baseline model provides a reference point against which classification performance can be judged. Motion energy extracted from the existing video recordings offers a low-cost control for movement-mediated coupling.

Operationalizing emotion via the dimensional valence-arousal model (Russell 1980) revealed differential sensitivity across affective dimensions that a combined categorical label would likely have obscured. Valence and arousal should therefore be treated as analytically distinct constructs and modeled as separate targets with dimension-specific refinements: alternative HRV-based markers beyond RMSSD, given its documented limitations (Choi et al. 2017), and higher-performing FER models than HSEmotion for valence. Validated self-report instruments such as the

Self-Assessment Manikin (Bradley and Lang 1994) may further strengthen interpretation, provided they complement rather than replace the physiological channels employed here.

The present analysis indicates that CO₂ does not confound the plant-valence relationship. Future studies could extend this by including volatile organic compound (VOC) measurements alongside CO₂ to assess whether other airborne chemical signals mediate or confound the plant-emotion relationship in ways that CO₂ alone cannot capture.

The finding that level-invariant and session-relative features consistently outranked absolute-level ones contributes to feature engineering in this domain and helps explain why session-normalized features tend to yield better performance. The descriptors in question are Hjorth mobility and complexity, both ratios and therefore scale-invariant, unlike Hjorth activity, a variance measure sharing the level dependence of the amplitude descriptors. Whether the pattern is physiological or instrumental can be subject to future research.

8 Conclusion

This study asked whether a human's emotional state, as measured through FER and HRV, can be predicted above chance from a plant's electrophysiological signal alone. A multimodal pilot study with 15 participants was conducted, combining heart rate variability and facial expression recognition into proxy indicators of arousal and valence, with ambient CO₂ recorded alongside. The plant signal was evaluated against these indicators since no self-reported ground truth was collected.

The answer to the study's main question is a qualified yes. Valence classification produced the clearest result (macro-F1 = 0.42, permutation $p = 0.005$) and recognized all three classes with comparable reliability, but at 43% accuracy against the approximately 33% expected by chance it is a statistical association across 284 windows and 15 participants, not a channel from which the state of an individual could be read at a practically useful rate. The five-class emotion label likewise reached significance ($p = 0.015$), yet its macro-F1 of 0.21 lies close to the uniform-chance expectation of approximately 0.20, and its 42% accuracy rests on the dominant neutral class; a label crossing valence with arousal cannot carry more than its weaker component, and arousal, resting solely on HRV, did not reach conventional significance ($p = 0.055$). H1 is therefore supported for valence, only nominally for the crossed emotion label, and not for arousal. The difference between the two dimensions should not be read as evidence about the dimensions themselves, since they were operationalized through proxies of unequal quality, a facial expression model on the one hand and a single HRV index on the other.

The secondary analysis tested a plausible alternative explanation for the valence finding, that ambient CO₂ rather than the plant tracked the participant's breathing and thus their emotional state. It does not hold: the plant-valence association survives controlling for CO₂ almost unchanged at both the raw and the session-normalized level. The plant therefore does not follow the air on the timescale at which the classification operates. The data provides no support for H2, which given the sample

size should be read as absence of evidence rather than as definitive proof.

These findings must be read against several structural limitations. All results are validated only against proxy labels; the sample is small, single-cohort, and recorded on a single day; the fixed stimulus order leaves session time as an unresolved confound; and no sham or shielding control rules out electromagnetic artifacts. The mechanism also remains open: excluding CO₂ removes an obvious mediating pathway without putting another in its place. Future work should prioritize a larger cohort with validated self-report ratings and arousal markers beyond RMSSD, and should replicate the present design across a broader population to determine whether the weak but statistically robust valence signal reflects a reproducible phenomenon.

A Appendix

A.1 Plant Feature Columns

Table 2 lists every column extracted from the plant voltage signal for one 30-second window, ordered by the three feature families of the Feature Engineering section. The 22 continuous columns are the ones entering the correlation analysis; the classification design matrix adds the six quality columns at the foot of the table, which describe the state of the electrode signal rather than its shape. Each continuous column appears in the dataset twice, once on its original scale and once as a per-session robust z-score carrying the suffix `_snorm` (Equation 11); the two representations are never mixed within one model.

Two implementation details apply throughout. The time-domain and signal-dynamics columns are computed after a zero-phase 46–54 Hz band-stop filter removes mains interference, which on the recorded sessions accounts for the majority of the raw signal’s total power and would otherwise dominate every amplitude statistic. The spectral columns are computed from the unfiltered signal so that the Welch spectrum stays faithful; for the same reason the band powers stop at 30 Hz and no band covering the mains frequency is formed.

Table 2. Plant feature columns: features extracted from the plant voltage signal per 30-second window. The 22 continuous columns additionally exist in a session-normalised variant with the suffix `_snorm`.

Column	Family	Description
<code>plant_mean</code>	Time	Mean voltage in the window
<code>plant_median</code>	Time	Median voltage
<code>plant_std</code>	Time	Standard deviation, retaining slow baseline drift
<code>plant_detrend_std</code>	Time	Standard deviation after linear detrending, i.e. fast fluctuation only
<code>plant_iqr</code>	Time	Interquartile range, $Q_{75} - Q_{25}$
<code>plant_min</code>	Time	Minimum voltage
<code>plant_max</code>	Time	Maximum voltage
<code>plant_range</code>	Time	Maximum minus minimum
<code>plant_rms</code>	Time	Root mean square
<code>plant_zcr</code>	Time	Zero-crossing rate about the window mean
<code>bp_0_1</code>	Spectral	Welch band power, 0–1 Hz
<code>bp_1_5</code>	Spectral	Welch band power, 1–5 Hz
<code>bp_5_15</code>	Spectral	Welch band power, 5–15 Hz
<code>bp_15_30</code>	Spectral	Welch band power, 15–30 Hz
<code>band_power_sum</code>	Spectral	Sum of the four band powers
<code>ratio_bp1_5_bp5_15</code>	Spectral	Ratio of the 1–5 Hz to the 5–15 Hz band power

Column	Family	Description
ratio_bp1_5_total	Spectral	Share of the 1–5 Hz band in the total band power
spectral_entropy	Spectral	Shannon entropy of the Welch spectrum, normalised to [0, 1]
hjorth_activity	Dynamics	Signal variance
hjorth_mobility	Dynamics	Mean-frequency proxy, from the variance of the first difference
hjorth_complexity	Dynamics	Frequency-change proxy, from the variance of the second difference
spike_rate	Dynamics	Share of samples deviating more than $3 \cdot 1.4826 \cdot \text{MAD}$ from the window median
plant_sat_pos_frac	Quality	Share of samples at the upper ADC rail
plant_sat_neg_frac	Quality	Share of samples at the lower ADC rail
plant_const_flag	Quality	All samples in the window identical
plant_quality	Quality	Verdict: ok, suspect_*, or dead_*
plant_dead_flag	Quality	Derived from plant_quality: signal unusable
plant_suspect_flag	Quality	Derived from plant_quality: signal usable but degraded

A.2 Labeling Equations

The equations below formalise the labeling procedure described in Section 4.4. Throughout, w indexes a non-overlapping 30-second window and the subscript s marks a statistic computed over all windows of the same session, which is what makes every threshold relative to the individual participant rather than to an absolute scale. Continuous per-window quantities are written in lower case (val_w , aro_w) and the discrete labels derived from them as L_w^{val} , L_w^{aro} and L_w^{emo} .

Valence from facial expression. Frames are sampled at 5 Hz. Let F_w be the sampled frames of window w in which HSEmotion detected a face, $v_i \in [-1, 1]$ the per-frame valence estimate and c_i the model's confidence in it. The window's valence is the confidence-weighted mean of those estimates:

$$\text{val}_w = \text{clip} \left(\frac{\sum_{i \in F_w} c_i v_i}{\sum_{j \in F_w} c_j}, -1, +1 \right) \quad (1)$$

Confidence weighting lets clearly visible faces dominate over frames the model itself rates as uncertain. A window is only labeled if a face was found in at least half of its sampled frames:

$$\text{cov}_w = \frac{|F_w|}{n_w}, \quad \text{window excluded if } \text{cov}_w < 0.5 \quad (2)$$

Arousal from heart-rate variability. Let $\text{RR}_1, \dots, \text{RR}_{N_w}$ be the artifact-corrected R-R intervals in milliseconds falling into window w . Two standard measures are computed from them, the root mean square of successive differences and the mean heart rate:

$$\text{RMSSD}_w = \sqrt{\frac{1}{N_w - 1} \sum_{i=1}^{N_w-1} (\text{RR}_{i+1} - \text{RR}_i)^2} \quad (3)$$

$$\overline{\text{HR}}_w = \frac{60000}{\frac{1}{N_w} \sum_{i=1}^{N_w} \text{RR}_i} \quad (4)$$

A window is excluded if it contains fewer than ten beats, or if more than 5 % of its beats had to be corrected. Both measures are then expressed as robust z-scores against the session's own centre and spread, using the median and the scaled median absolute deviation rather than mean and standard deviation so that individual outlying windows cannot move the reference:

$$z_s(x) = \frac{x - \text{med}_s(x)}{1.4826 \cdot \text{MAD}_s(x)} \quad (5)$$

Because RMSSD falls with physiological activation while heart rate rises with it, RMSSD is inverted before the two are averaged into one arousal index:

$$\text{aro}_w = \frac{1}{2} \left(-z_s(\text{RMSSD}_w) + z_s(\overline{\text{HR}}_w) \right) \quad (6)$$

Deadbands. Each axis receives a neutral deadband whose half-width is derived from the spread of that axis' own values within the session:

$$\tau_{\text{val}} = k \cdot 1.4826 \cdot \text{MAD}_s(\text{val}), \quad \tau_{\text{aro}} = k \cdot 1.4826 \cdot \text{MAD}_s(\text{aro}) \quad (7)$$

The multiplier is fixed at $k = 0.5$ on both axes, so the deadband spans roughly one robust standard deviation in total. It was selected on a pilot session as the largest value that still left a substantial share of windows directional: larger values collapse the dataset into the neutral class, smaller ones label deviations that lie within the participant's ordinary resting fluctuation.

Discretisation. Both axes are discretised by the same rule, each against its own median and its own deadband:

$$L_w^{\text{val}} = \begin{cases} \text{positive} & \text{if } \text{val}_w > \text{med}_s(\text{val}) + \tau_{\text{val}} \\ \text{negative} & \text{if } \text{val}_w < \text{med}_s(\text{val}) - \tau_{\text{val}} \\ \text{neutral} & \text{otherwise} \end{cases} \quad (8)$$

$$L_w^{\text{aro}} = \begin{cases} \text{activated} & \text{if } \text{aro}_w > \text{med}_s(\text{aro}) + \tau_{\text{aro}} \\ \text{calm} & \text{if } \text{aro}_w < \text{med}_s(\text{aro}) - \tau_{\text{aro}} \\ \text{neutral} & \text{otherwise} \end{cases} \quad (9)$$

Crossing into the emotion label. The two axis labels are combined into the final five-class label following the quadrant structure of the circumplex model (Figure 1). A window only receives a directional label if both axes are directional:

$$L_w^{\text{emo}} = \begin{cases} \text{neutral} & \text{if } L_w^{\text{val}} = \text{neutral} \vee L_w^{\text{aro}} = \text{neutral} \\ (L_w^{\text{val}}, L_w^{\text{aro}}) & \text{otherwise} \end{cases} \quad (10)$$

so that L_w^{emo} takes one of the five values *negative calm*, *negative activated*, *positive calm*, *positive activated*, or *neutral*.

Session normalisation of the plant features. The `_snorm` representation of Table 2 applies the same robust standardisation as Equation 5, but to each plant feature x rather than to an HRV measure:

$$x_w^{\text{snorm}} = \frac{x_w - \text{med}_s(x)}{1.4826 \cdot \text{MAD}_s(x)} \quad (11)$$

This removes the constant offsets that electrode placement, plant state, and ambient conditions introduce between recordings, while preserving the within-session variation the classifier is meant to exploit.

A.3 Example Session Record

Figure 6 documents one complete recording, session 2026-07-06_19-51_proband_15, as the labeling algorithm of Section 4.4 sees it. The same processing was applied unchanged to every session; this one is shown because it is a full-length recording whose windows carry labels from all five classes. The upper four panels hold the quantities the algorithm reads, each plotted against the session median it is judged relative to, and the two lower panels the labels it produces, so reading the figure downwards reproduces the order of the labeling procedure. The plant voltage and the ambient CO₂ trace share the fourth panel because the correlation analysis of Section 5 tests exactly this pair against each other.

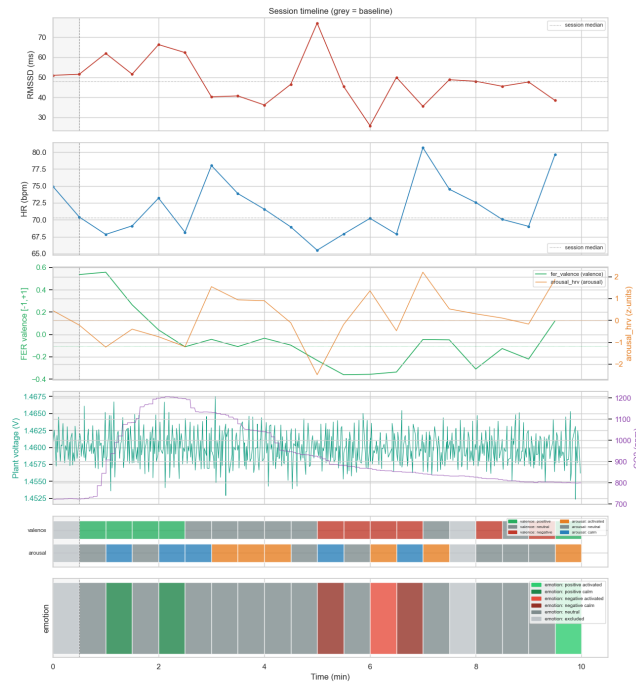


Fig. 6. Example session record: the full ten-minute recording of session 2026-07-06_19-51_proband_15, plotted on the non-overlapping 30-second window grid. From top: RMSSD; mean heart rate; the two quantities that decide the labels, FER valence on the left axis (green) and the HRV arousal index in robust z-units on the right axis (orange); the plant voltage (teal, left axis) with ambient CO₂ overlaid (purple, right axis); the two per-axis labels; and the crossed five-class emotion label. Dotted horizontal lines mark the session medians each window is judged against (Equation 8, Equation 9). The shaded band at the left and the dashed vertical line mark the 30-second baseline preceding the video. In the two label rows, light grey marks windows excluded rather than labeled, following Equation 2. The plant trace is decimated to 1 Hz for display only; the features of Table 2 are computed on the undecimated signal.

A.4 Experimental Setup and Live Dashboard

The two figures below document the physical side of the study. Figure 7 shows the sensor setup described in the Sensor Setup and Field Work sections, and Figure 8 the real-time component of Section 4.7. Both are collected here rather than in the method chapter because neither is required to follow the argument of the paper: the live component demonstrates the trained models rather than contributing evidence for the hypotheses.



Fig. 7. Physical setup of the experiment: the *Kalanchoe* plant with its three leaf electrodes, the AD8232 and the ESP32 on the breadboard, and the laptop the participant faces during a session. The photograph was taken while the reduced demonstration configuration was running, so the CO₂ sensor is not attached and the screen shows the Phenomena variant of the dashboard rather than the recording software.

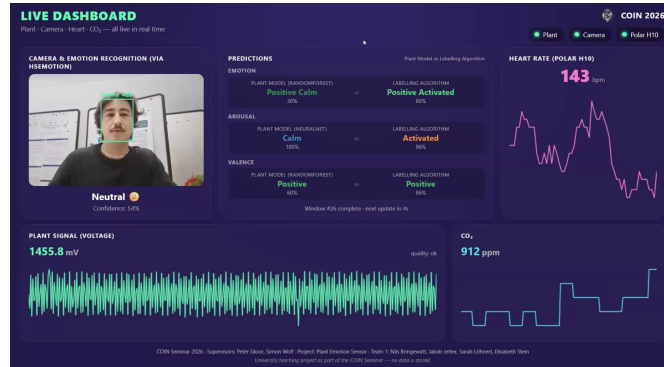


Fig. 8. The live dashboard: the camera panel shows the HSEmotion result for the current frame, the centre column contrasts the plant-model prediction with the labeling algorithm's proxy label for each of the three axes, and the remaining panels show the live plant voltage, heart rate, and CO₂ streams. The footer of the prediction column reports the window counter and the time until the next update.

A.5 Confusion Matrices

The three figures below hold the confusion matrices behind the per-class scores quoted in Section 5, one per target variable. Each figure covers all six (model, representation) combinations rather than only the winning one, so the per-class behaviour of a combination can be compared against the alternatives it was selected over. The counts are pooled over the five outer folds of the nested cross-validation, which is why every panel sums to the 284 windows of the dataset: each window is held out, and therefore predicted, exactly once.

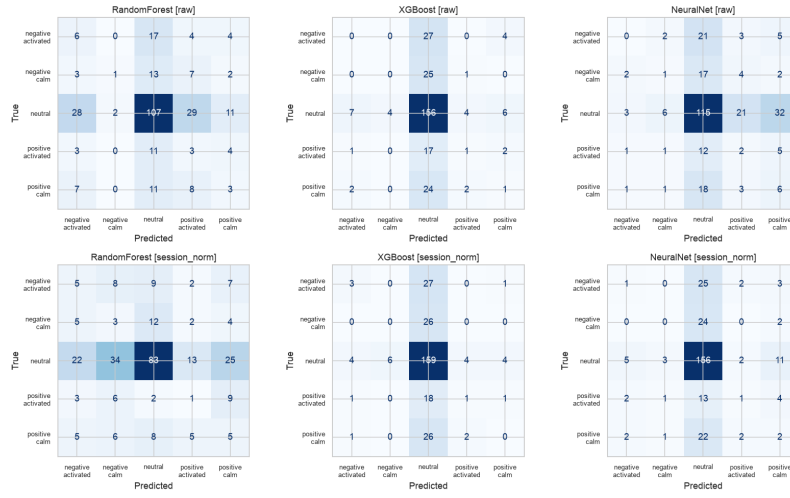


Fig. 9. Confusion matrices for the target variable emotion: pooled over the five outer folds of the nested cross-validation, for all six (model, representation) combinations. Rows hold the proxy label and columns the model's prediction, so the diagonal counts the correctly classified windows and each panel sums to the 284 windows of the dataset.

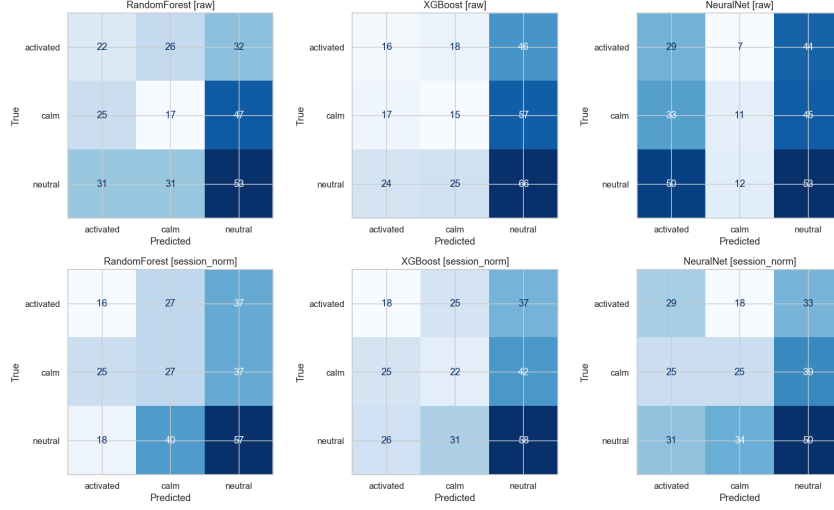


Fig. 10. Confusion matrices for the target variable arousal: pooled over the five outer folds of the nested cross-validation, for all six (model, representation) combinations. Rows hold the proxy label and columns the model's prediction, so the diagonal counts the correctly classified windows and each panel sums to the 284 windows of the dataset.

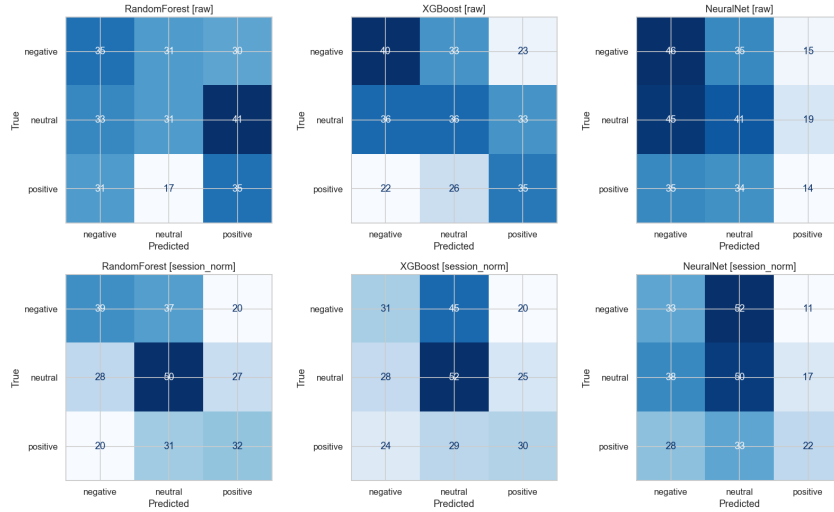


Fig. 11. Confusion matrices for the target variable valence: pooled over the five outer folds of the nested cross-validation, for all six (model, representation) combinations. Rows hold the proxy label and columns the model's prediction, so the diagonal counts the correctly classified windows and each panel sums to the 284 windows of the dataset.

A.6 Permutation Tests

The three figures below document the permutation tests behind the p-values reported in Section 5, one per target variable. Each test repeatedly shuffles the labels and re-runs the complete participant-grouped five-fold loop with the winning architecture held fixed, building the distribution of macro-F1 this architecture attains when no systematic relationship between the plant features and the label is left. The p-value is the share of those 200 permutations that matched or exceeded the value observed on the real labels. In each figure the grey histogram is that null distribution, the solid red line the observed value, and the dashed blue line the pooled nested cross-validation estimate of Table 1, which is the more conservative of the two because it re-tunes the model inside every fold instead of holding one architecture fixed.

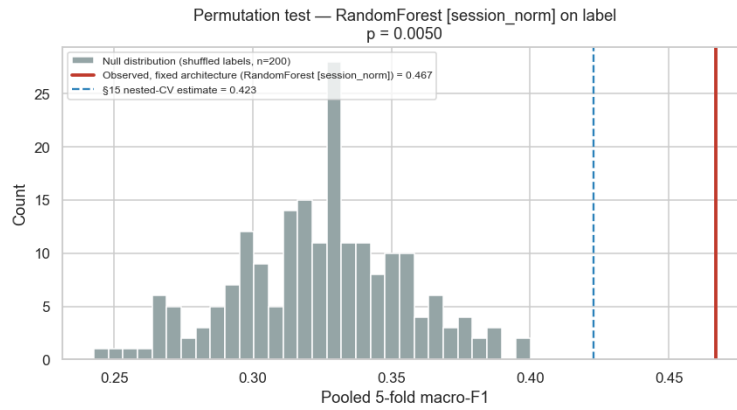


Fig. 12. Permutation test for the target variable valence: null distribution of the pooled macro-F1 over 200 label permutations for Random Forest on the session-normalized features, against the observed value. None of the 200 permutations reached it ($p = 0.005$).

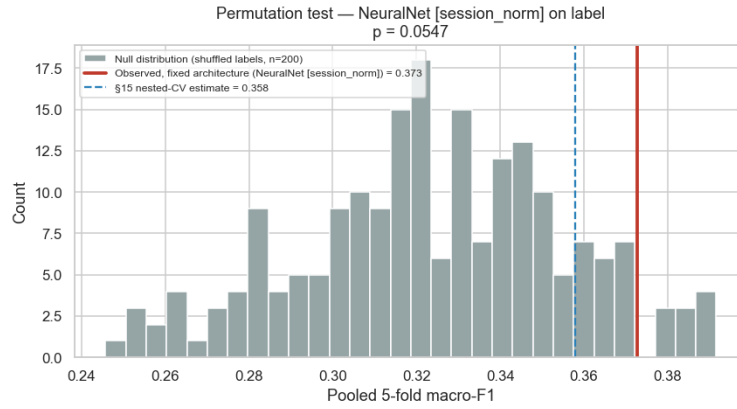


Fig. 13. Permutation test for the target variable arousal: null distribution of the pooled macro-F1 over 200 label permutations for the Neural Network on the session-normalized features, against the observed value. Ten of the 200 permutations matched or exceeded it ($p = 0.055$), so this target misses the conventional significance threshold.

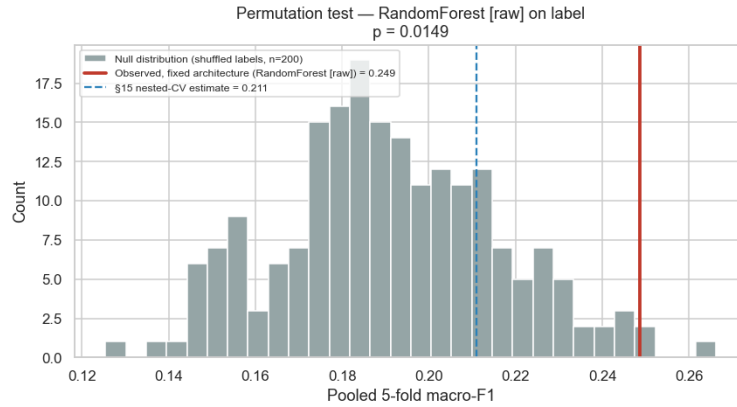


Fig. 14. Permutation test for the target variable emotion: null distribution of the pooled macro-F1 over 200 label permutations for Random Forest on the raw features, against the observed value. Two of the 200 permutations matched or exceeded it ($p = 0.015$).

A.7 Feature Importances

The three figures below document the feature rankings quoted in Section 5, one per target variable. Only the two tree-based architectures expose `feature_importances_`, which is why the arousal ranking is read from the tree models rather than from the multilayer perceptron that won that target. Each figure pairs the two tree models on the representation the winning combination uses, so a ranking that holds for both can be told apart from one that only a single model produces. The importances come from the models tuned on the full dataset, the only place the pipeline exposes them; that pass is separate from the nested cross-validation behind every reported score in Table 1, so these rankings are diagnostic rather than a performance metric.

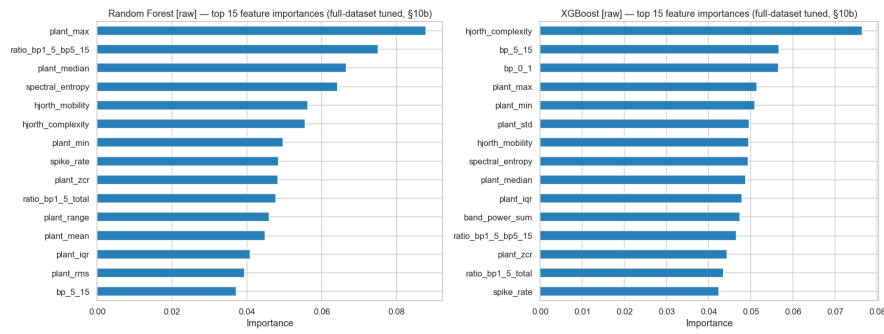


Fig. 15. Feature importances for the target variable emotion: the top 15 features of the two tree-based models on the raw representation, which is the representation the winning Random Forest uses.

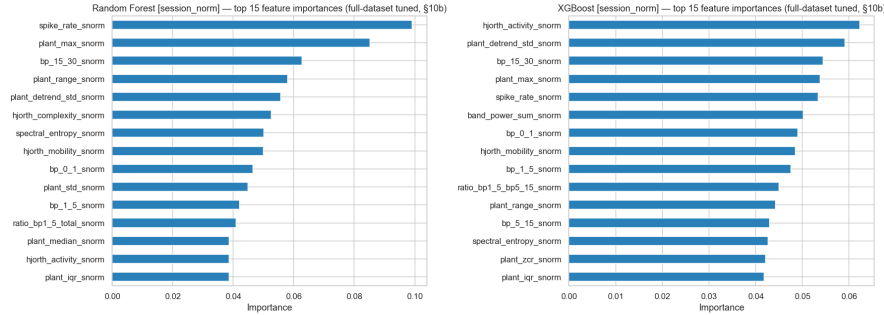


Fig. 16. Feature importances for the target variable arousal: the top 15 features of the two tree-based models on the session-normalized representation. The winning multilayer perceptron exposes no feature importances, so the ranking is read from the tree models on the same representation.

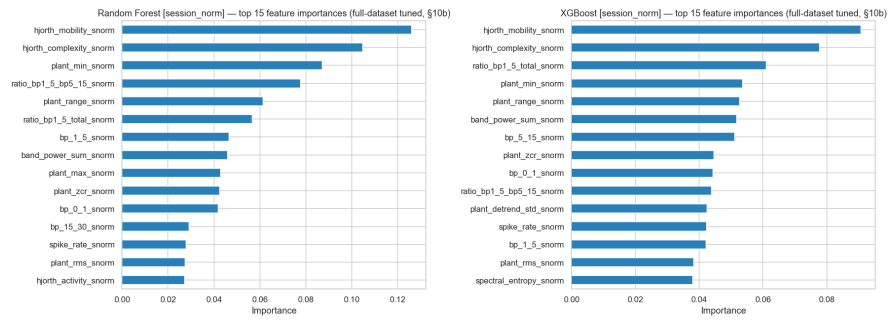


Fig. 17. Feature importances for the target variable valence: the top 15 features of the two tree-based models on the session-normalized representation, which is the representation the winning Random Forest uses.

A.8 CO₂ Confound Analysis

The two figures below document the correlation analysis of Section 5, one per variant: Figure 18 the raw analysis, whose correlations carry both within- and between-session variation, and Figure 19 the session-normalized analysis, in which every signal is centred on its own session so that only within-session variation remains. Both are read the same way. The left panel is the confounding triangle, drawn with a solid edge where the relationship is present by the criterion of Section 4.8 and a dashed one where it is not, and with the plant–valence edge labelled before and after CO₂ is controlled for. The right panel resolves that edge per feature, plotting the absolute correlation with valence before (purple) and after (vermillion) removing CO₂, so a feature whose association is carried by CO₂ appears as a long line pointing toward zero.

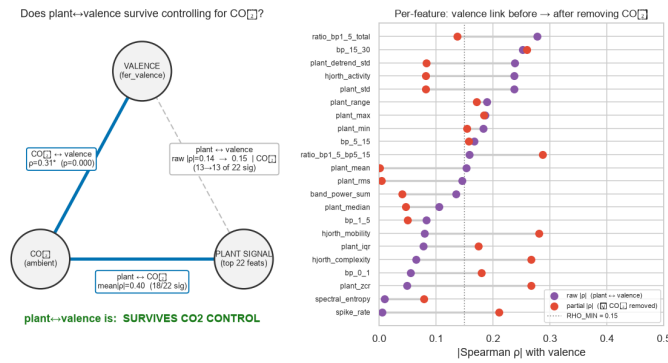


Fig. 18. Confounding triangle and partial correlations, raw analysis: the plant, valence and CO₂ triangle, and the per-feature partial correlation with valence before and after removing CO₂. Both within- and between-session variation contribute.

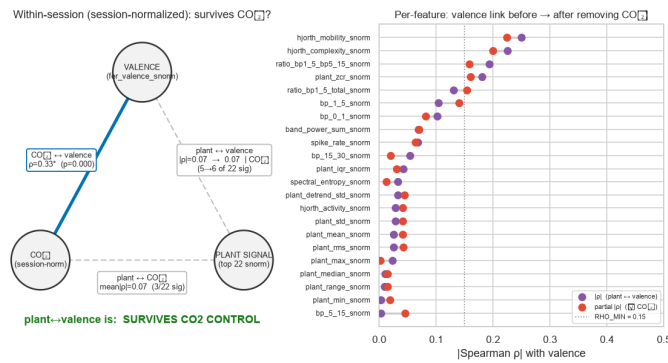


Fig. 19. Confounding triangle and partial correlations, session-normalized analysis: the same three edges and the same per-feature comparison after the plant features, CO₂ and valence have each been centred on their own session, so that only within-session variation remains.

A.9 Code Repository

Every component described in this paper is hosted at

https://github.com/JakobJetter/COIN2026_Team-1_Plant-Emotion-Detector

The repository is organised along the pipeline stages of Figure 2. The `arduino/` directory holds the ESP32 firmware that samples the AD8232 and the CO₂ sensor and streams both over WebSocket. The notebooks under `notebooks/` then run in numeric order: `01_init_recording` records one session into a shared event log and a video file, `02_emotion_labelling` derives the per-window HRV, FER and plant features and applies the labeling rule of Section 4.4, `03_dataset_creation` runs that labeling over every session and merges the result into the consolidated dataset, the three `04_*_ml_pipeline` notebooks train and compare the classifiers for emotion, arousal and valence respectively, and `05_Correlation_Analysis` performs the CO₂ confound analysis reported in Section 5. The live dashboard of Section 4.7 is implemented in `emotion_live_dashboard.py` under `phaenomena/`. The consolidated dataset and the correlation result table ship under `data/datasets/`, so the modelling and correlation notebooks can be re-run without repeating the recordings.

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